

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: INJONGO | Context: Ethiopian Intercity/Ride-Hail Commuter Cohort (Amharic-Primary)

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The 23-slot final ontology [\[Q118, Q119, Q144\]](#) includes structural fits for deployment needs (CITY_OR_PROVINCE, MONEY, CURRENCY, DATE, TIME, BANK_NAME), and merging of low-frequency types is principled [\[Q51, Q52, Q55\]](#). However, several decision-rule mismatches arise for the Ethiopian deployment: (a) DATE/TIME boundary rules are documented [\[Q153, Q154\]](#) using Gregorian conventions only — no accommodation for Ethiopian calendar (13 months, Pagume intercalary) or 6-hour offset clock; (b) the 40-intent inventory excludes intercity bus, seat-class, fare-tier, and rebooking categories required for Sky Bus/Selam Bus/Ride [\[Q39, DATASET-D7, D8\]](#); (c) dataset inspection shows Amharic DATE annotations dominated by relative expressions, suggesting the operative decision rule for Ethiopian month names is undertested [\[DATASET-D15, D16, D28\]](#). These are structural-validity gaps. A redeeming factor: Ethiopian month names did appear as DATE slots [\[DATASET-D2, D6\]](#), so the ontology can host them in principle.

ID	Evidence
Q39	"The source data for our multilingual benchmark is from the CLINC English dataset—an intent detection with 150 intent classes across 10 domains (but without slot annotation), we extracted 40 intents from five most suitable domains to the African contexts: Banking (e.g. "transfer", "pay bill"), Home (e.g. "play music", "calendar update"), Kitchen and Dining (e.g. "recipe", "confirm reservation"), Travel (e.g. "exchange rate", "book flight"), and Utility (e.g. " (p.3)
Q51	"We decided to exclude slot entities appearing less than 500 times across all languages (after MUSIC_GENRE in the figure)." (p.4)
Q52	"Consequently, nine infrequent slots from NATIONALITY through PLUG_TYPE were eliminated." (p.4)
Q55	"This merging process resulted in a reduction from 34 to 23 slot types." (p.5)
Q118	"The "Original Slot Type"are used during the dataset annotation phase, which contained 34 slot types." (p.13)
Q119	"After merging similar or low-frequency types during data preprocessing in Section 3.3, it was reduced to 23 distinct slot types as shown in the "Final Merged Type" column." (p.13)
Q144	"# Named Entities Types to Identify ACCOUNT_TYPE, ARTIST_NAME, BANK_NAME, BILL_TYPE, CALENDAR_EVENT, CITY_OR_PROVINCE, COUNTRY, CURRENCY, DATE, DISH_OR_FOOD, HOTEL_NAME, LANGUAGE_NAME, MEAL_PERIOD, MONEY, MUSIC_GENRE, NUMBER, PAYMENT_COMPANY, PERSONAL_NAME, PLACE_NAME, RESTAURANT_NAME, SHOPPING_ITEM, SONG_NAME, TIME" (p.19)
Q153	"Anything that is less than one day should be annotated as TIME and not DATE, as seen in the above examples." (p.21)
Q154	"Anything that is more than one day must be annotated as DATE and not time, as seen above" (p.21)
D2	target: "CITY_OR_PROVINCE: ድሬዳዋ \$\$\$ CITY_OR_PROVINCE: ባህርዳር \$\$\$ DATE: መስከረም አምስት" — Slot annotations for Dire Dawa, Bahir Dar (Ethiopian cities), and Meskerem (Ethiopian calendar month)
D6	ከሃምሌ 4 እስከ 6 የሰዓት መኪና ኪራይ ማግኘት እችላለሁ?" / target: "DATE: ከሃምሌ 4 እስከ 6 — "Can I find a car rental from Hamle 4 to 6?" — Hamle is the 11th month of Ethiopian calendar
D7	ወደ ቱርክ በፍጥነት የምድረስበት የበረራ አማራጭ ንገረኝ። / target: "COUNTRY: ቱርክ — "Tell me a flight option to get to Turkey quickly." — international destination, no Ethiopian intercity booking
D8	ወደ ሱዳን የበረራ ትኬት ያስፈልገኛል። / target: "COUNTRY: ሱዳን — "I need a flight ticket to Sudan." — international destination, not intercity bus
D15	የዚህ ወር መከፈል ያለባቸውን የፍጆታ ዝርዝር አሳየኝ። / target: "DATE: የዚህ ወር — "Show me the utility list to be paid this month." — relative date expression, not calendar month name
D16	ከሚመጣው ወር ጀምሮ ለ3 ወር ቶዮታ መኪና መክራየት እፈልጋለሁ። / target: "DATE: ከሚመጣው ወር ጀምሮ ለ3 ወር — "I want to rent a Toyota car starting next month for 3 months." — relative date, no named calendar month

D28	በዚህ ሳምንት ምንም ቀጠሮ እንዲኖረኝ አልፈልግም።" / target: "DATE: በዚህ ሳምንት — Relative week reference, no Ethiopian calendar month
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Strengths

- 23 slot types including travel-relevant CITY_OR_PROVINCE, MONEY, CURRENCY, DATE, TIME [Q144]
- Detailed boundary rules for ambiguous cases (TIME vs DATE, BANK_NAME word inclusion) [Q152, Q153, Q154]
- Ontology can in principle host Ge'ez month names — confirmed empirically [DATASET-D2, D6]
- Merging process reduced ambiguity from 34 to 23 types [Q55, Q119]

ID	Evidence
Q55	"This merging process resulted in a reduction from 34 to 23 slot types." (p.5)
Q119	"After merging similar or low-frequency types during data preprocessing in Section 3.3, it was reduced to 23 distinct slot types as shown in the "Final Merged Type" column." (p.13)
Q144	"# Named Entities Types to Identify ACCOUNT_TYPE, ARTIST_NAME, BANK_NAME, BILL_TYPE, CALENDAR_EVENT, CITY_OR_PROVINCE, COUNTRY, CURRENCY, DATE, DISH_OR_FOOD, HOTEL_NAME, LANGUAGE_NAME, MEAL_PERIOD, MONEY, MUSIC_GENRE, NUMBER, PAYMENT_COMPANY, PERSONAL_NAME, PLACE_NAME, RESTAURANT_NAME, SHOPPING_ITEM, SONG_NAME, TIME" (p.19)
Q152	"When annotating Bank names, you do not need to include "bank" unless it is attached to the bank name, like seen above, with Ecobank." (p.21)
Q153	"Anything that is less than one day should be annotated as TIME and not DATE, as seen in the above examples." (p.21)
Q154	"Anything that is more than one day must be annotated as DATE and not time, as seen above" (p.21)
D2	target: "CITY_OR_PROVINCE: ድሬዳዋ \$\$\$ CITY_OR_PROVINCE: ባህርዳር \$\$\$ DATE: መስከረም አምስት" — Slot annotations for Dire Dawa, Bahir Dar (Ethiopian cities), and Meskerem (Ethiopian calendar month)
D6	ከሃምሌ 4 እስከ 6 ያሉት መኪና ኪራይ ማግኘት እችላለሁ?" / target: "DATE: ከሃምሌ 4 እስከ 6 — "Can I find a car rental from Hamle 4 to 6?" — Hamle is the 11th month of Ethiopian calendar

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Most slot types are regionally relevant (cities, money, banks). However, the intent ontology lacks bus-operator-specific categories required for the deployment [Q39, DATASET-D30].

OO-2: Missing: intercity bus intents, seat-class/fare-tier intents, rebooking-on-disruption intent. DATE rule does not encode Ethiopian calendar month set; TIME rule does not encode Ethiopian clock convention [Q153, Q154].

OO-3: DATE/TIME annotation conventions are implicitly Gregorian/international as documented [Q153, Q154]; this encodes a non-Ethiopian temporal convention as default, with no documented annotator instruction for Ge'ez calendar or 6-hour-offset clock.

OO-4: Stakeholder-driven taxonomy redesign is recommended for the intent layer (add bus-specific intents) and for the DATE/TIME decision rules (specify Ge'ez calendar handling and Ethiopian clock disambiguation).

OO-5: Documented issues: missing bus intents harm content validity; Gregorian-only DATE/TIME decision rules harm structural validity for Ethiopian temporal expressions.

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Q39	“The source data for our multilingual benchmark is from the CLINC English dataset—an intent detection with 150 intent classes across 10 domains (but without slot annotation), we extracted 40 intents from five most suitable domains to the African contexts: Banking (e.g. "transfer", "pay bill"), Home (e.g. "play music", "calendar update"), Kitchen and Dining (e.g. "recipe", "confirm reservation"), Travel (e.g. "exchange rate", "book flight"), and Utility (e.g." (p.3)
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Q153	“Anything that is less than one day should be annotated as TIME and not DATE, as seen in the above examples.” (p.21)
Q154	“Anything that is more than one day must be annotated as DATE and not time, as seen above” (p.21)
D30	no intercity corridors, no seat class, no bus operator, no fare tier present — All Amharic travel utterances use air travel framing

Information Gaps

- Whether the annotation guide (full Appendix) includes any Ge'ez calendar examples beyond the verbatim quotes available

Requires Expert Verification

- Whether 'CALENDAR_EVENT' would accommodate Timkat/Meskel as labels — taxonomy intent unclear from documentation

Remediation

Gap 1: DATE/TIME annotation rules implicitly Gregorian/international; no documented decision rule for Ge'ez calendar or Ethiopian 6-hour-offset clock; no intent categories for bus-specific flows

Recommendation: Author a Ge'ez-calendar and Ethiopian-clock annotation addendum to the INJONGO guide, specifying acceptable surface forms for each Ge'ez month, intercalary Pagume handling, and disambiguation rules for ambiguous TIME expressions; add bus-specific intents to the operative ontology for downstream evaluation.